

Structure, Composition and Circulation of the Atmosphere



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L31: 01 April 2024

L32: 03 April 2024

L33: 05 April 2024

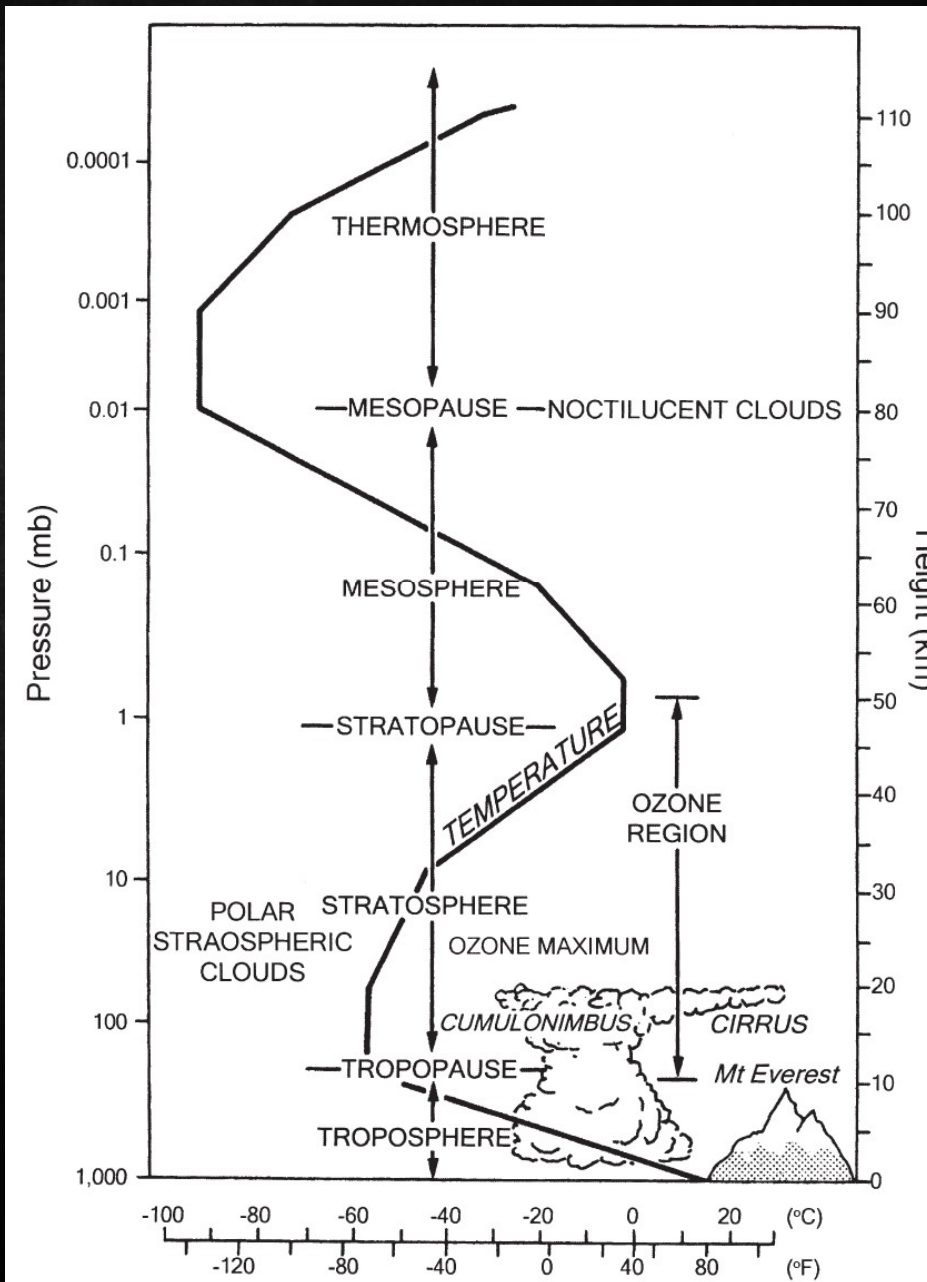
What is Atmosphere?

- Earth's atmosphere is unique. No other planet in our solar system has an atmosphere with the exact mixture of gases or the heat and moisture conditions necessary to sustain terrestrial life.
- The atmosphere envelops the earth to a thickness of only 1 per cent of the earth's radius. Dry air, by volume, is more than 99 per cent composed of nitrogen and oxygen and lie within 30 km of the Earth's surface.
- It had evolved to its present form and composition at least 400 million years ago by which time a considerable vegetation cover had developed the land.
- At its base, the atmosphere rests on the ocean surface which, at present, covers some 70 per cent of the surface of the globe.
- Although air and water share somewhat similar physical properties, they differ in one important respect – air is compressible, water incompressible.

THE LAYERING OF THE ATMOSPHERE

- The atmosphere can be divided conveniently into a number of rather well-marked horizontal layers, mainly on the basis of temperature.
- There are three relatively warm layers (near the surface; between 50 and 60 km; and above about 120 km) separated by two relatively cold layers (between 10 and 30 km; and 80 and 100 km).
- The evidence for this structure comes from regular rawinsonde (radar wind-sounding) balloons, radio wave investigations, and, more recently, from rocket flights and satellite sounding systems.

Vertical structure of the Atmosphere



1. Troposphere

- The troposphere is characterized by a decrease in temperature with respect to height, at a rate of about 6.5°C per kilometer, up to a height of about 10 km.
- The troposphere is the lowest layer of Earth's atmosphere and is the site of all weather on Earth.
- The troposphere is bonded on the top by a layer of air called the tropopause, which separates the troposphere from the stratosphere, and on bottom by the surface of the Earth.
- It is the zone where weather phenomena and atmospheric turbulence are most marked, and it contains 75 per cent of the total molecular or gaseous mass of the atmosphere and virtually all the water vapour and aerosols.
- Throughout this layer, there is a general decrease of temperature with height at a mean rate of about $6.5^{\circ}\text{C}/\text{km}$.

1. Troposphere.....Contd.

- The decrease in temperature occurs because air is compressible and its density decreases with height, allowing rising air to expand and thereby cool.
- The troposphere contains 99% of the water vapor in the atmosphere.
- In addition, turbulent heat transfer from the surface mainly heats the atmosphere, not direct absorption of radiation.
- The troposphere is capped in most places by a temperature inversion level (i.e. a layer of relatively warm air above a colder layer) and in others by a zone that is isothermal with height.
- The troposphere thus remains to a large extent self-contained, because the inversion acts as a 'lid' that effectively limits convection.
- This inversion level or weather ceiling is called the tropopause .
- Its height is not constant in either space or time. It seems that the height of the tropopause at any point is correlated with sea-level temperature and pressure, which are in turn related to the factors of latitude, season and daily changes in surface pressure.

1. Troposphere.....Contd.

- There are marked variations in the altitude of the tropopause with latitude, from about 16 km at the equator, where there is strong heating and vertical convective turbulence, to only 8 km at the poles.

2. Stratosphere

- The stratosphere extends upward from the tropopause to about 50 km and accounts for about 10 per cent of the atmospheric mass.
- Although the stratosphere contains much of the total atmospheric ozone (it reaches a peak density at approximately 22 km), maximum temperatures associated with the absorption of the sun's ultraviolet radiation by ozone occur at the stratopause, where they may exceed 0°C.
- The air density is much lower here, so even limited absorption produces a high temperature rise.
- Temperatures increase fairly generally with height in summer, with the coldest air at the equatorial tropopause. In winter, the structure is more complex with very low temperatures, averaging -80°C, at the equatorial tropopause, which is highest at this season.

2. Stratosphere.....Contd.

- Similar low temperatures are found in the middle stratosphere at high latitudes, whereas over 50 - 60°N there is a marked warm region with nearly isothermal conditions at about -45 to -50°C.
- Marked seasonal changes of temperature affect the stratosphere.

3. Mesosphere

- Above the stratopause, average temperatures decrease to a minimum of about -133°C (140 K) at around 90 km.
- This layer is commonly termed the mesosphere, although as yet there is no universal terminology for the upper atmospheric layers.
- Pressure is very low in the mesosphere, decreasing from about 1 mb at 50 km to 0.01 mb at 90 km. Above 80 km, temperatures again begin rising with height and this inversion is referred to as the mesopause.
- Molecular oxygen and ozone absorption bands contribute to heating around 85 km altitude.
- It is in this region that noctilucent clouds (a luminous cloud occasionally seen at night in summer in high latitudes) are observed on summer 'nights' over high latitudes.

4. Thermosphere

- Atmospheric densities are extremely low above the mesopause, although the tenuous atmosphere still effects drag on space vehicles above 250 km.
- The lower portion of the thermosphere is composed mainly of nitrogen (N_2) and oxygen in molecular (O_2) and atomic (O) forms, whereas above 200 km atomic oxygen predominates over nitrogen (N_2 and N).
- Temperatures rise with height, owing to the absorption of extreme ultraviolet radiation (0.125 to 0.205 μm) by molecular and atomic oxygen, probably approaching 800 to 1200 K at 350 km, but these temperatures are essentially theoretical.

COMPOSITION (CONSTITUENTS) OF THE ATMOSPHERE

1. Primary gases: Air is a mechanical mixture of gases, not a chemical compound. Dry air, by volume, is more than 99 per cent composed of nitrogen and oxygen. These gases are mixed in remarkably constant proportions up to about 100 km altitude. Yet, despite their predominance, these gases are of little climatic importance.

2. Greenhouse gases (CO_2 , CH_4 , N_2O , O_3 , CFCs, Hydrogenated halocarbons, H_2O)

In spite of their relative scarcity, the so-called *greenhouse gases* play a crucial role in the thermodynamics of the atmosphere. They trap radiation emitted by the earth, thereby producing the *greenhouse effect*. Moreover, the concentrations of these trace gases are strongly affected by human (i.e. anthropogenic) activities:

3. Reactive gas species

In addition to the greenhouse gases, important reactive gas species are produced by the cycles of sulphur, nitrogen and chlorine (Reactive nitrogen species: nitric oxide (NO) and nitrogen dioxide (NO₂); Reactive sulphur species: sulphur dioxide (SO₂) and reduced sulphur (H₂S); acid deposition includes both acid rain and snow (wet deposition) and dry deposition of particulates). These play key roles in acid precipitation and in ozone destruction. Sources of these species are Nitrogen species, Sulphur species and Acid deposition.

4. Aerosols

Aerosols are suspended particles of sea-salt, mineral dust (particularly silicates), organic matter and smoke. Aerosols enter the atmosphere from a variety of natural and anthropogenic sources. Some originate as particles - soil grains and mineral dust from dry surfaces, carbon soot from coal fires and biomass burning, and volcanic dust.

Aerosol (<5 μm radius) production sources and typical concentrations ($\mu\text{g m}^{-3}$).

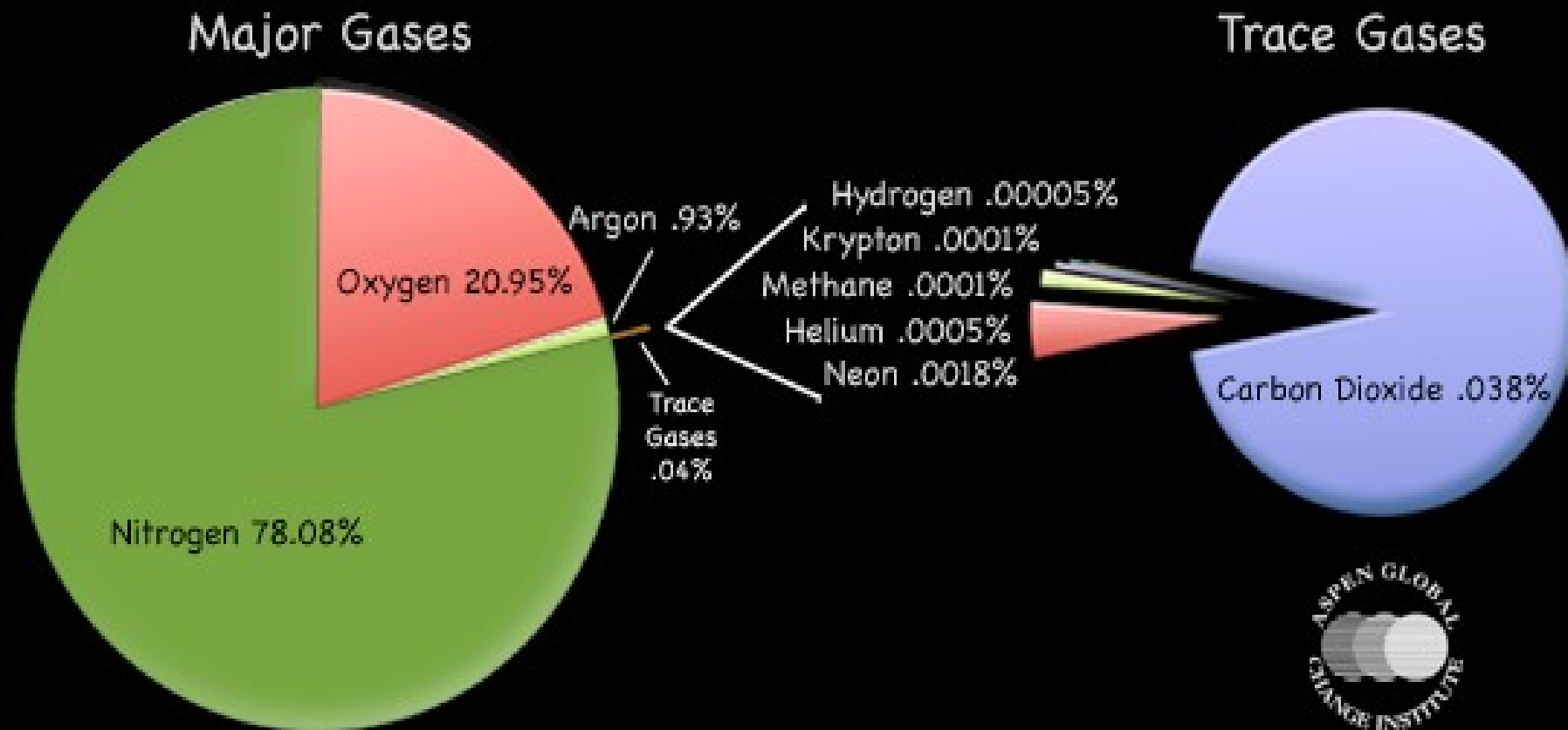
A. Natural Sources	Production
1. Primary Production	
Sea Salt	2300
Mineral Particles	900-1500
Volcanics	20
Forest Fires & Biological Debris	50
2. Secondary production (gas particle):	
Sulphur from H_2S	70
Nitrates from NO_x	22
Converted plant hydrocarbons	25
Total Natural	~3687

Aerosol (<5 μm radius) production sources and typical concentrations ($\mu\text{g m}^{-3}$).....contd.

B. Anthropogenic	Production
1. Primary Production	
Mineral Particles	0-600
Industrial Dust	50
Combustion (Black Carbon)	10
Organic Carbon	50
2. Secondary Production (Gas Particle):	
Sulphur from SO_2	140
Nitrates from NO_x	30
Biomass combustion (Organic)	20
Total Anthropogenic	~600

Composition of the Atmosphere

Composition of the Atmosphere



Average composition of the dry atmosphere below 30 km

Constituent	Percent by volume	Concentration in parts per million (PPM)
Nitrogen (N ₂)	78.084	780,840.0
Oxygen (O ₂)	20.946	209,460.0
Argon (Ar)	0.934	9,340.0
Carbon dioxide (CO ₂)	0.038	380.0
Neon (Ne)	0.00182	18.2
Helium (He)	0.000524	5.24
Methane (CH ₄)	0.00015	1.5
Krypton (Kr)	0.000114	1.14
Hydrogen (H ₂)	0.00005	0.5

Atmospheric Gas Laws

Atmospheric gases obey a few simple laws in response to changes in pressure and temperature.

1. Boyle's Law, states that, at a constant temperature, the volume (V) of a mass of gas varies inversely as its pressure (P), i.e.

$$P = \frac{k_1}{V}$$

or, $P_1V_1 = P_2V_2 = K_1$
Where, k_1 is a constant.

2. Charles's Law, states that, at a constant pressure, volume varies directly with absolute temperature (T) measured in degrees Kelvin.

$$V = k_2T$$

Or, $V_1 / T_1 = V_2 / T_2 = k_2$

Atmospheric Gas Laws.....Contd.

3. Avogadro's Law, (hypothesized in 1811) states that at a constant temperature and pressure, the volume occupied by an ideal gas is directly proportional to the number of molecules of the gas present. This gives rise to the molar volume of a gas, which at STP (273.15 K, 1 atm) is about 22.4 L. The relation is given by:

$$V \propto n, \text{ or}$$
$$\frac{V_1}{n_1} = \frac{V_2}{n_2}$$

where n is the number of molecules of gas (or the number of moles of gas).

These laws imply that the qualities of pressure, temperature and volume are completely interdependent, such that any change in one of them will cause a compensating change to occur in one, or both, of the remainder.

Atmospheric Gas Laws.....Contd.

The gas laws (Boyle's and Charles's Laws) may be combined to give the following relationship:

$$PV = RmT$$

where m = mass of air, and R = Universal gas constant for dry air ($287 \text{ J kg}^{-1} \text{ K}^{-1}$, or, $8.3144598 \text{ (kPa}\cdot\text{L)/(mol}\cdot\text{K)}$). If m and T are held fixed, we obtain Boyle's Law; if m and P are held fixed, we obtain Charles's Law. Since it is convenient to use density, ρ (= mass/volume), rather than volume when studying the atmosphere, we can rewrite the equation in the form known as the equation of state:

$$P = R\rho T$$

With the addition of Avogadro's law, the combined gas law develops into the ideal gas law:

$$PV = nRT$$

An equivalent formulation of this law is: $PV = nk_B T$

Where k_B is the Boltzmann constant ($1.381 \times 10^{-23} \text{ J}\cdot\text{K}^{-1}$ in SI units).

Numerical-01: If a 17.5 mL sample of gas is initially at a pressure of 4500 atm, what will be the volume of the gas when the pressure becomes 1500 atm, assuming the amount of gas and temperature remain constant?

$$\begin{aligned} V_2 &= \frac{P_1 \cdot V_1}{P_2} \\ &= \frac{4.500 \text{ atm} \cdot 17.50 \text{ mL}}{1.500 \text{ atm}} \\ &= 52.50 \text{ mL} \end{aligned}$$

Numerical-02: If a sample of carbon dioxide in a pump has a volume of 20.5 mL at 40°C, what will be the volume of carbon dioxide in the pump when the temperature is increased to 65°C, assuming the amount of gas and pressure remain constant?

$$\begin{aligned} V_2 &= \frac{V_1 \cdot T_2}{T_1} \\ &= \frac{20.5\text{mL} \cdot (60 + 273.15\text{K})}{40 + 273.15\text{K}} \\ &= 22.1\text{mL} \end{aligned}$$

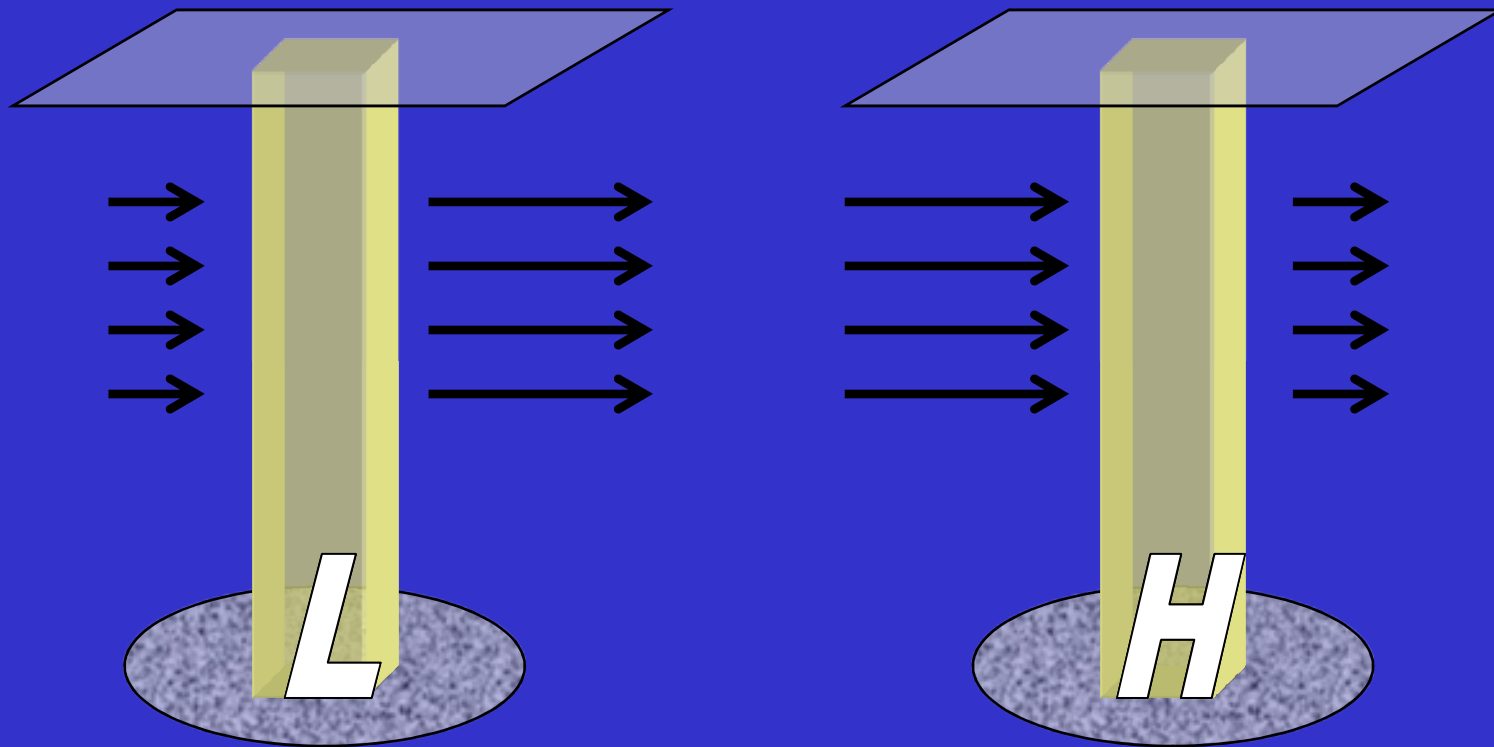
Horizontal & Vertical Motion of the Atmosphere

Air Pressure and Air Density

Density = Mass / Volume

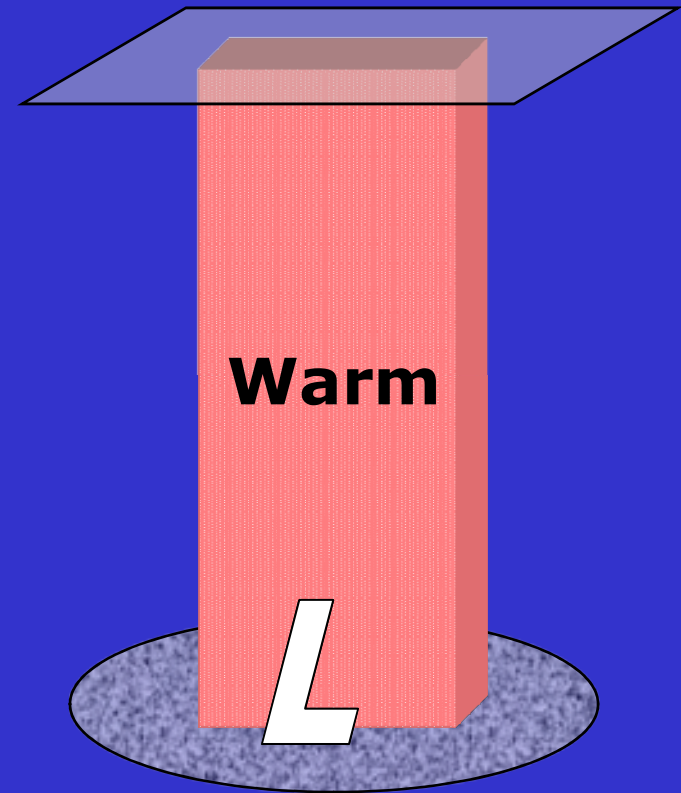
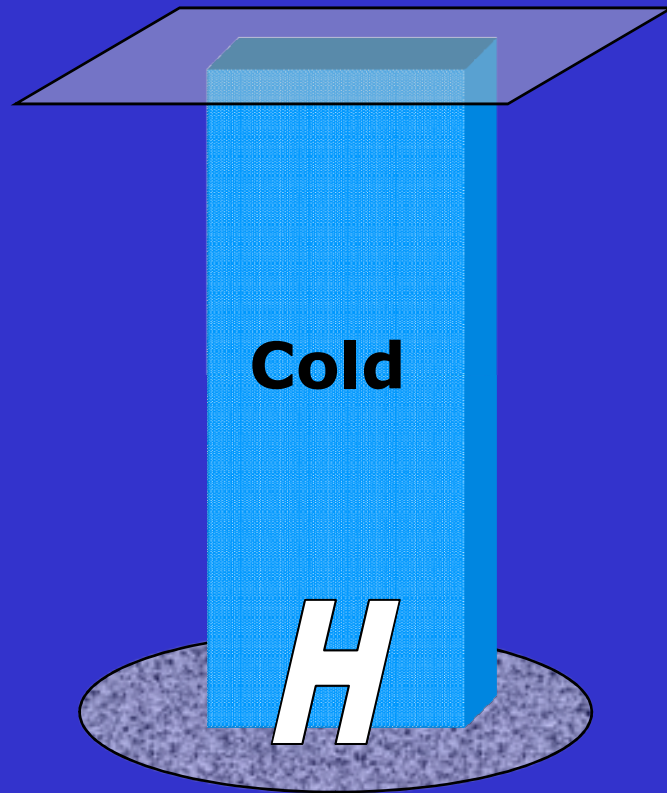
Pressure = Force / Area

Changing Pressure - Winds



Take more out, than put in – decrease pressure
Put more in, than take out – increase pressure

Changing Pressure - Temperature

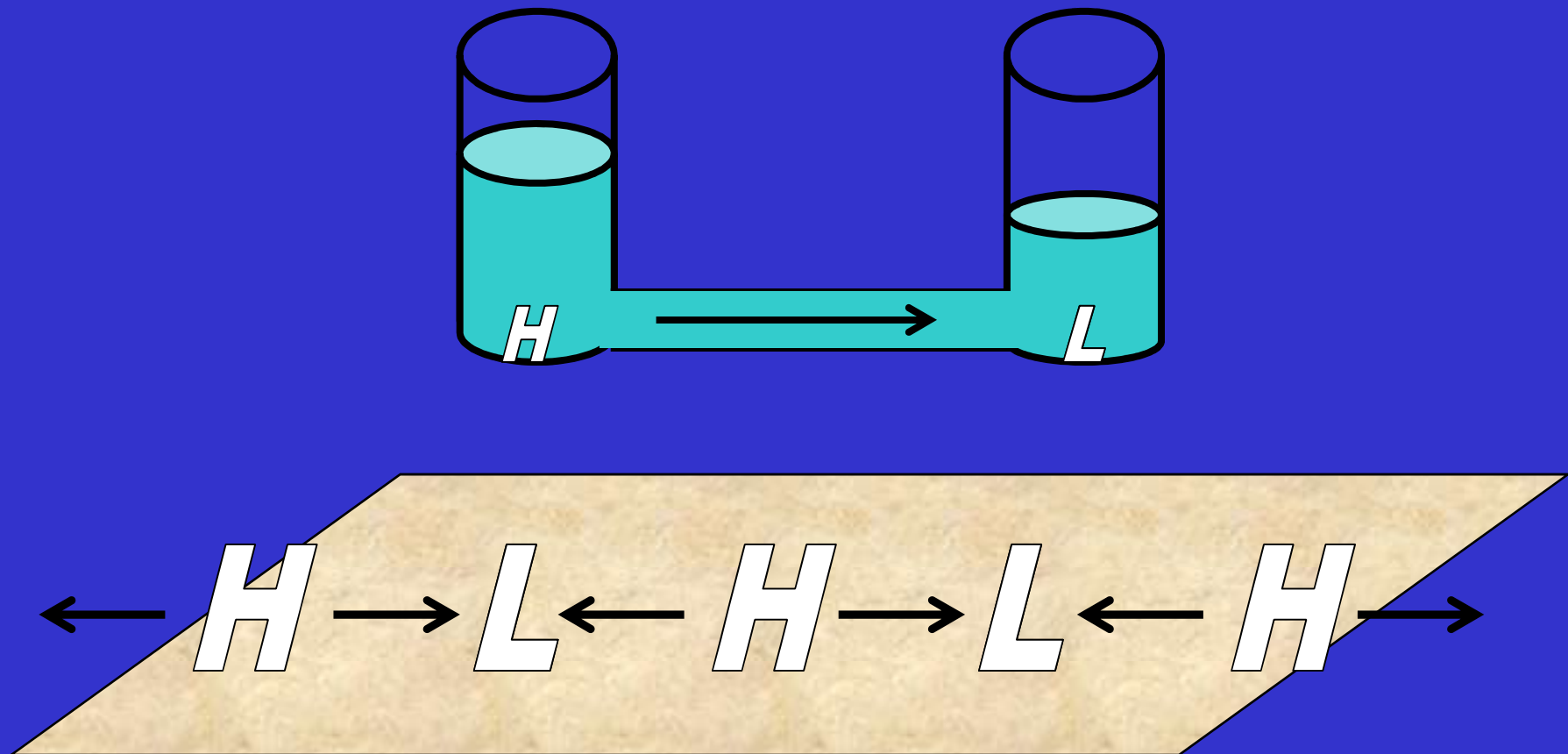


Coldest column = highest pressure **

Warmest column = lowest pressure **

Pressure Differences Create Wind

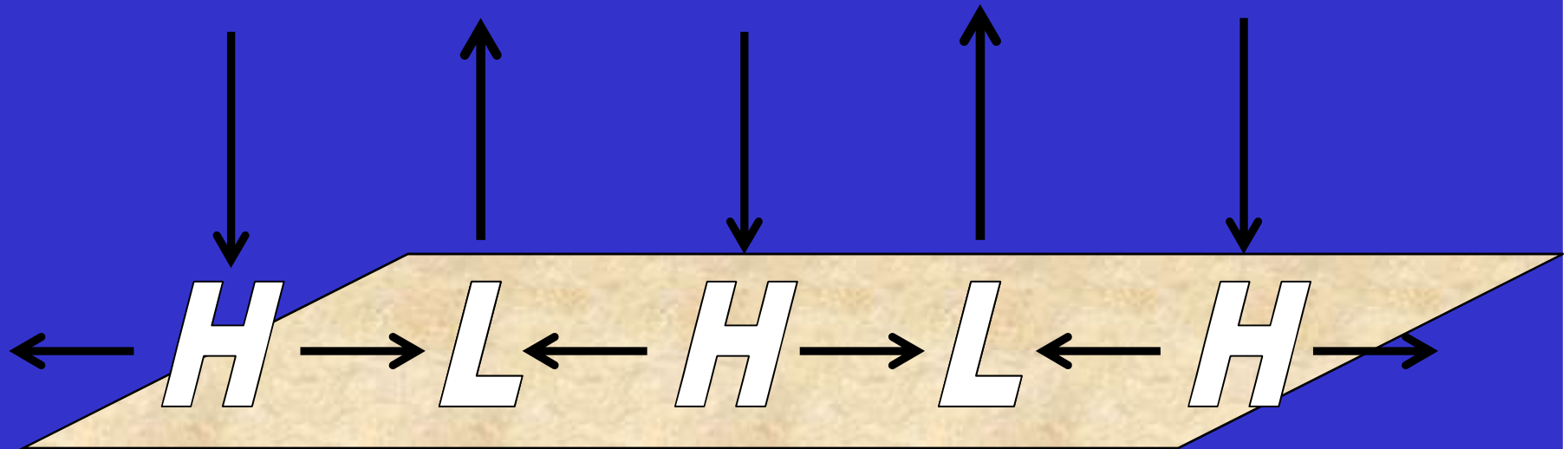
Horizontal Motion of the Wind



Air moves from higher toward lower pressure

Vertical motions also occur

Vertical Motion of the Wind



Air “converges” at lows, and rises.

Air “diverges” from highs, and sinks.

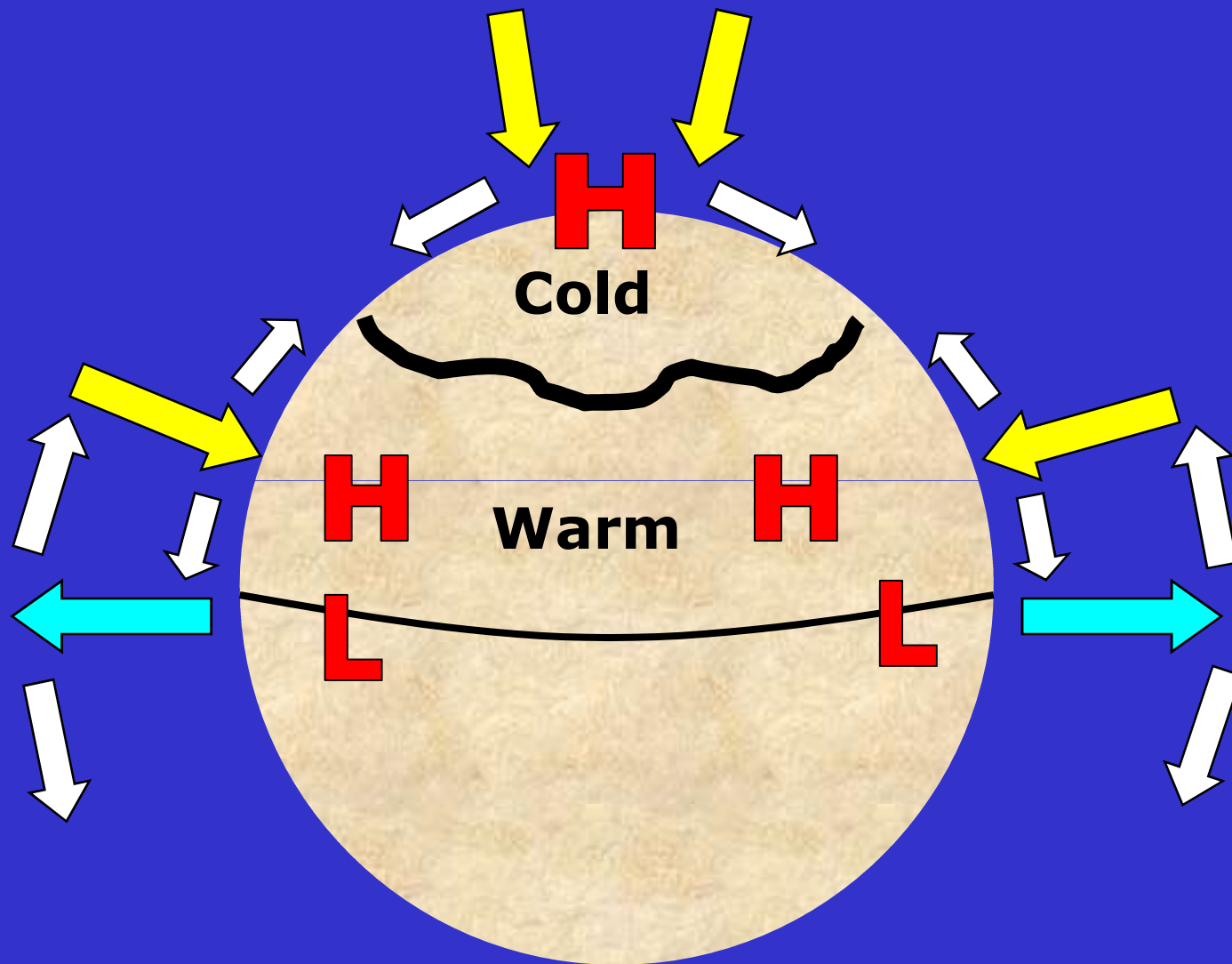
Rising Air near **L**ows

- Rising air cools; water vapor in the air condenses to form clouds/precipitation
- Lows tend to bring cloudy, wet weather

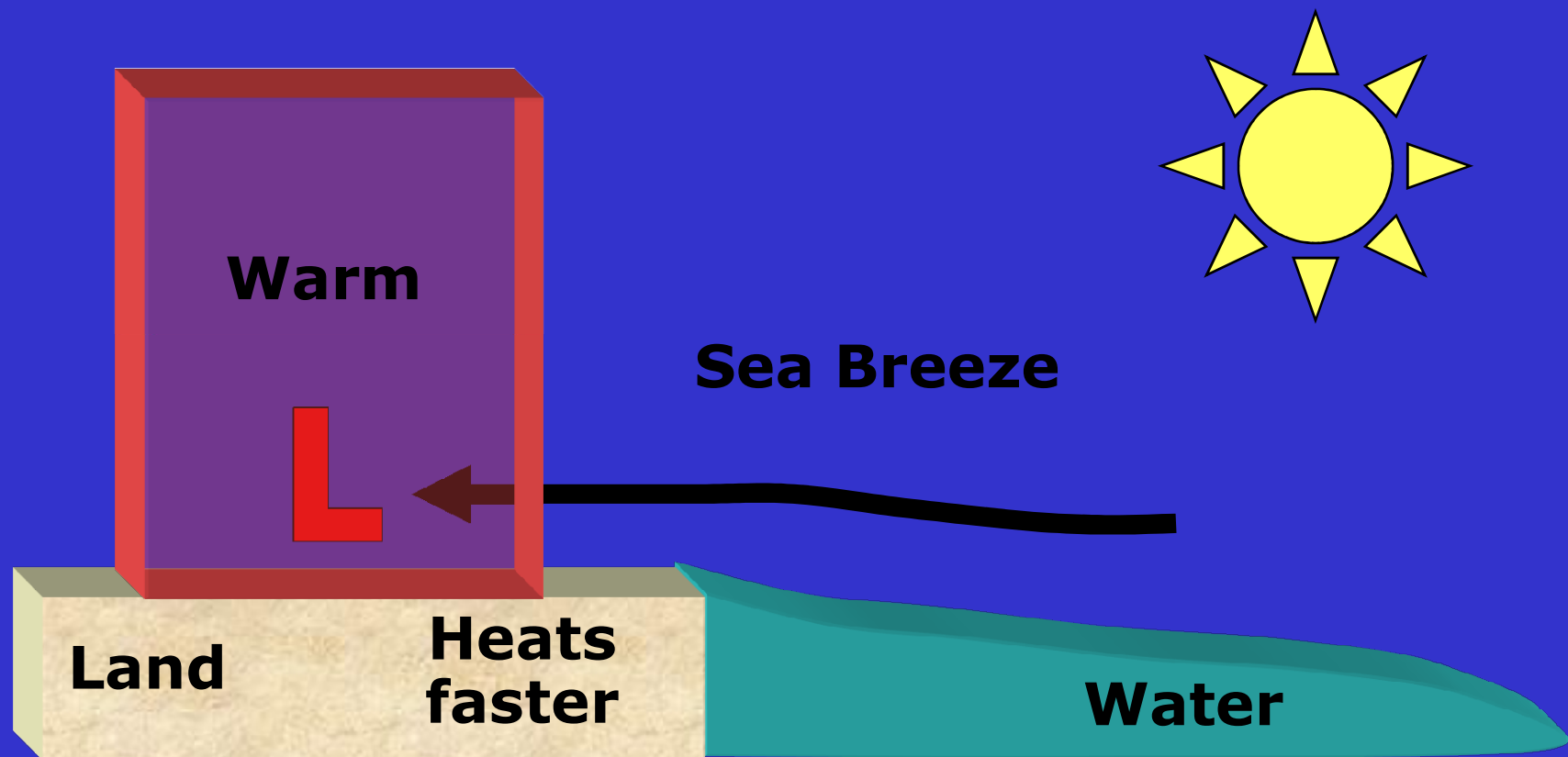
Sinking air near **H**ighs

- Sinking air warms and dries out.
- Highs tend to bring fair, dry weather.

General Circulation



Local Winds



Uneven heating working on a smaller scale

General Circulation of the Atmosphere

Atmospheric Circulation

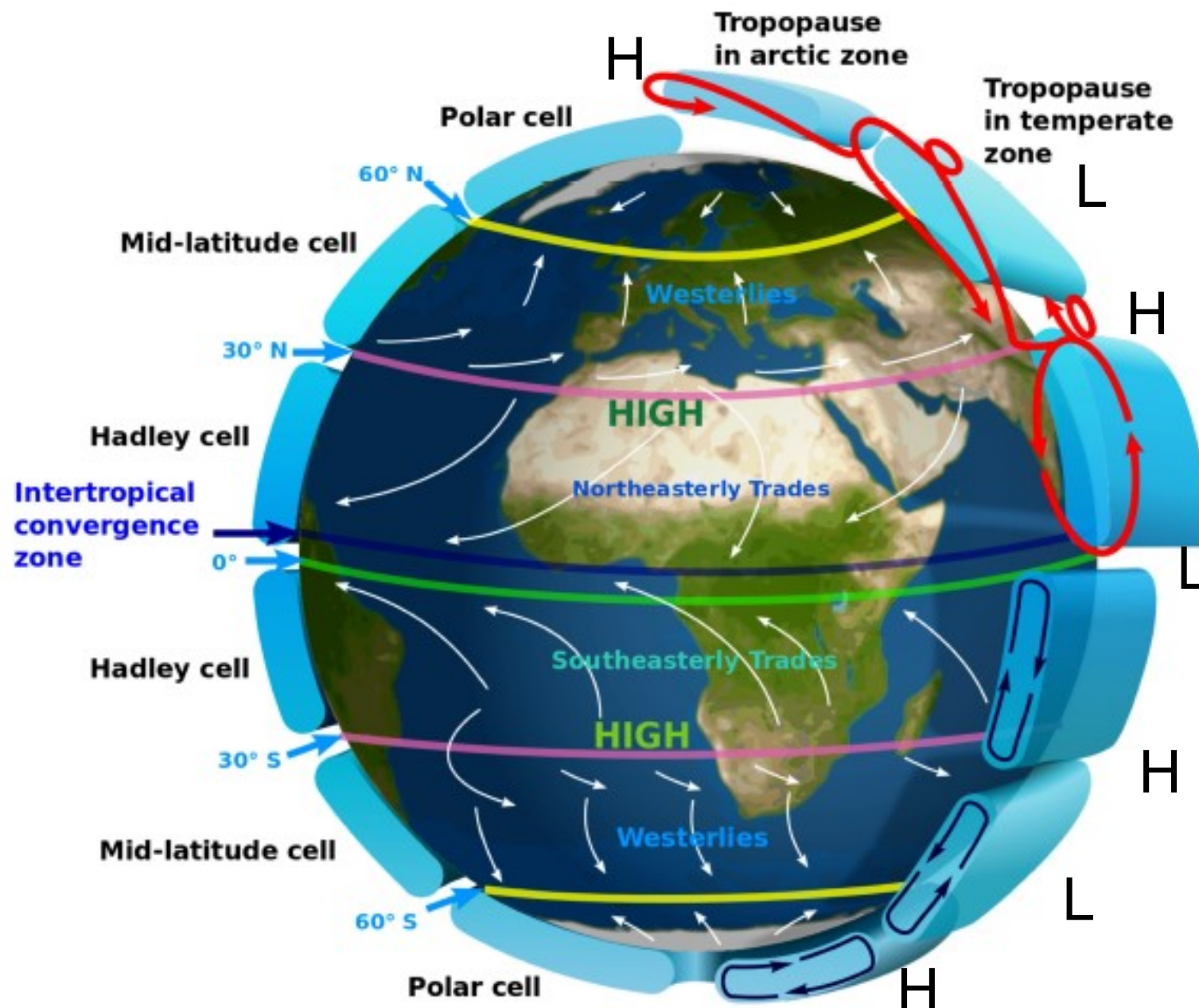
- Atmospheric circulation is the large-scale movement of air and together with ocean circulation is the means by which thermal energy is redistributed on the surface of the Earth.
- The Earth's atmospheric circulation varies from year to year, but the large-scale structure of its circulation remains fairly constant.
- The movement of air on a global scale transfers heat from the equator to the poles. At the same time, the large-scale movement of air generates latitudinal belts of prevailing easterly and westerly winds. The movement of air is known as the general circulation of the atmosphere

Factors for General Circulation of the Atmosphere

The pattern of planetary winds largely depends on:

- Latitudinal variation of atmospheric heating
- The emergence of pressure belts
- The migration of belts following the apparent path of the sun
- The distribution of continents and oceans
- The rotation of the earth

Three-Cell Model

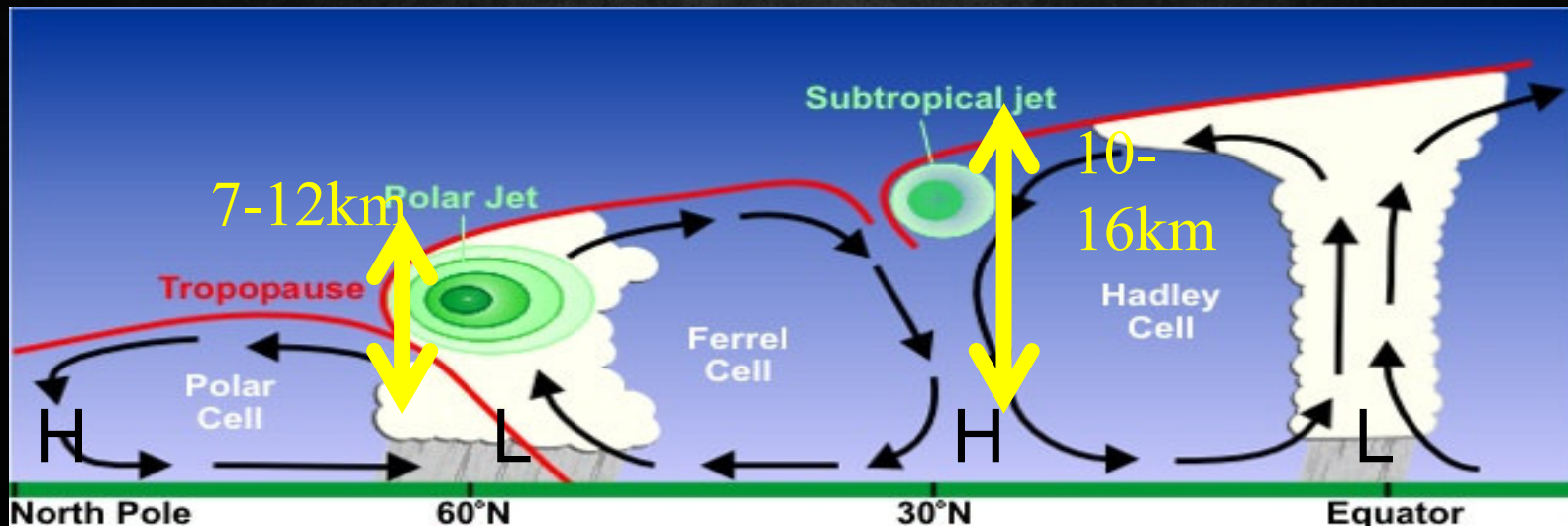


INTER-TROPICAL CONVERGENCE ZONE (ITCZ)

- **IN THE NORTHERN HEMISPHERE:** The winds that blow to the equatorial low pressure belt are called the North East Trade Winds.
- The line along which they converge (meet) is called the **INTERTROPICAL CONVERGENCE ZONE**. This is often abbreviated to **ITCZ**.
- **IN THE SOUTHERN HEMISPHERE:** The winds that blow to the equatorial low pressure belt are called the South East Trade Winds.
- The line along which they converge (meet) is called the **ITCZ**.

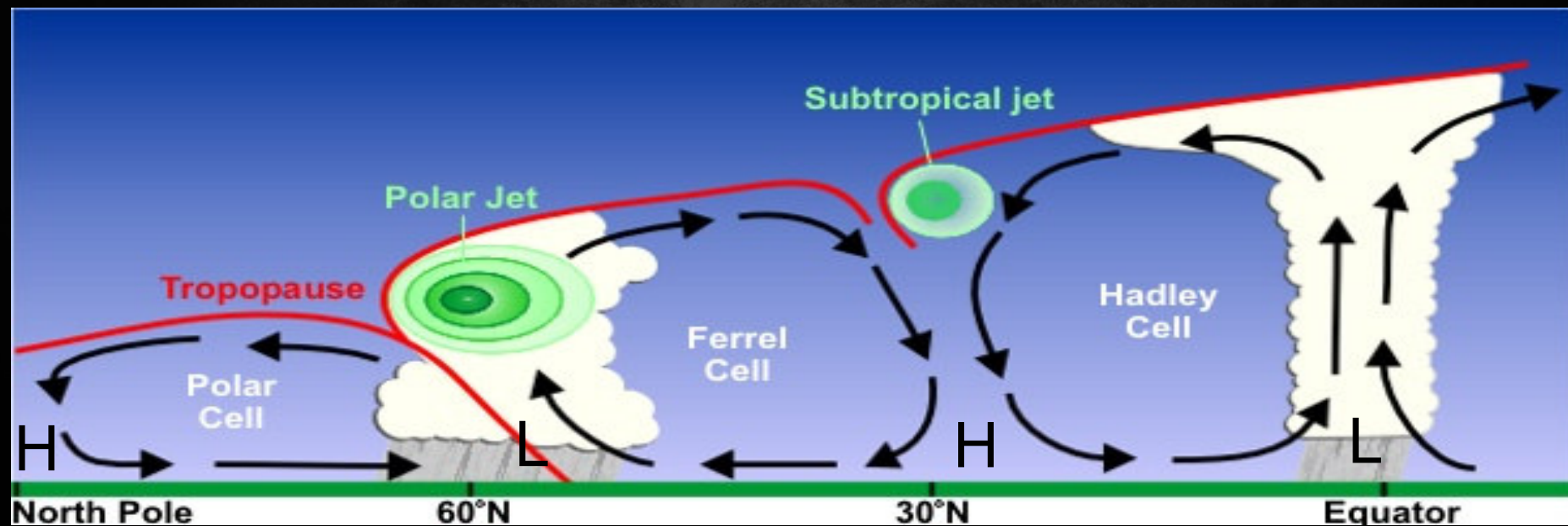
1. FORMATION OF THE HADLEY CELL

- Insolation in tropical areas causes warm air to rise and spread polewards, carrying heat energy.
- Air cools and begins to fall at about 30°N and 30°S of Equator. Cooled air returns to the Equator. This circulation of air is caused by solar heating.
- Heat energy is transferred from the Equator to subtropical latitudes.
- This circular motion is called the HADLEY CELL.



2. FORMATION OF THE POLAR CELL

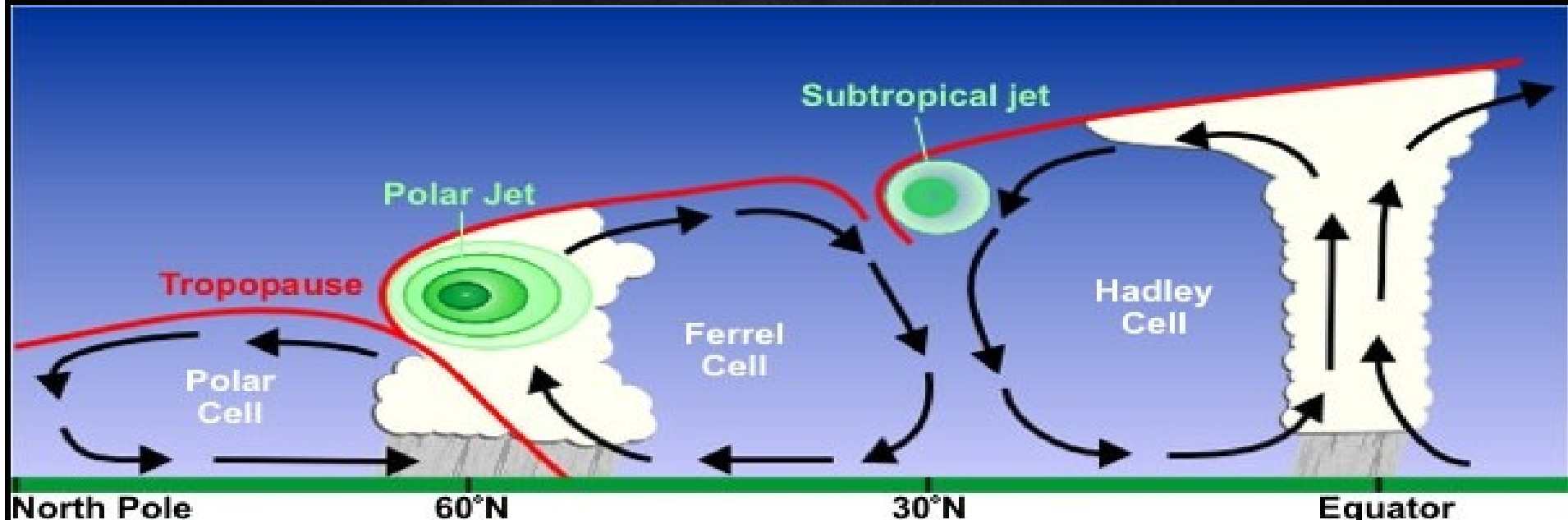
- Intensely cold, dense air sinks at the poles, then blows as surface winds towards the Equator.
- At about 60°N and 60°S, the cold polar air is warmed in contact with the earth's surface. This warmed air rises and returns polewards, carrying heat energy.
- This circular motion is called the POLAR CELL.



3. FORMATION OF THE FERREL CELL

- The Hadley Cell is driven by differences in heat energy at the Equator.
- As the air in the Hadley Cell falls at about 30°N and 30°S , it pulls the air beside it down as well, due to friction.
- The Polar Cell is driven by differences in heat energy. Cold polar air falls and spreads towards the Equator.
- As the air in the Polar Cell rises at about 60°N and 60°S , it pulls the air beside it up as well, due to friction
- Unlike the Hadley and Polar Cells, the Ferrel Cell is not driven by differences in heat energy
- The Ferrel Cell is caused by friction where air is in contact with the other two cells.
- The Hadley Cell drags air down at about 30°N and 30°S .

3. FORMATION OF THE FERREL CELL



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Jet streams

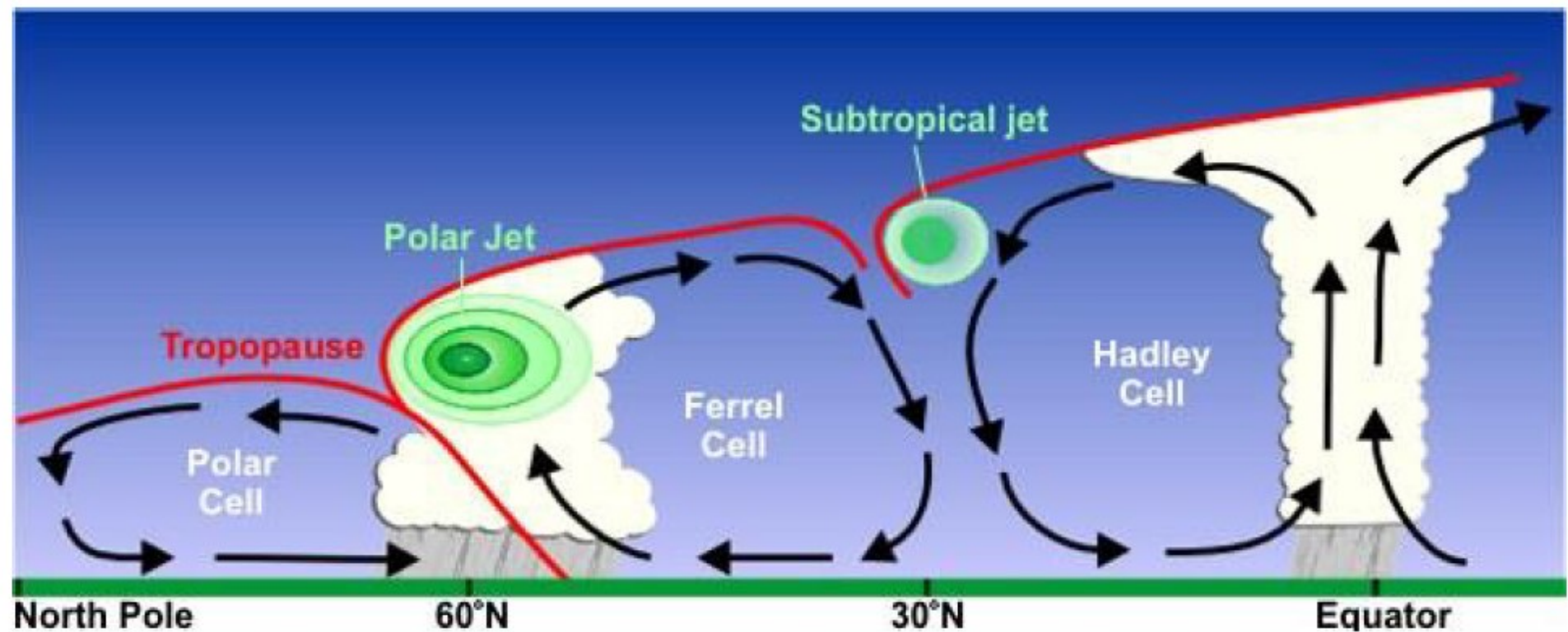
- Jet streams are relatively narrow bands of strong wind in the upper levels of the atmosphere. The winds blow from west to east in jet streams but the flow often shifts to the north and south.
- Jet streams follow the boundaries between hot and cold air.
- There are two jet streams: the subtropical jet and the polar jet, located right under the tropopause.
- The figure shows the average position of the jet streams in the Northern Hemisphere in the winter, as well as their relation to the tropopause.
- The jet streams flowing from west to east. The subtropical jet stream is located near 30° latitude about 13 km up, above the tropical high.
- The polar jet stream is located near the polar front about 10 km up, near 50° to 60° latitude.

Jet streams.....Contd.

- The difference in height of these jets is due to their location at the tropopause, and the fact that the tropopause is found higher in tropical regions than in polar regions due to average layer temperature differences of the troposphere underneath.
- The troposphere is thinner in polar regions than in tropical regions due to colder, denser air at the poles.
- Jet streams are thousands of kilometers in length, spanning across continents, while being relatively narrow, and span a few hundred kilometers wide. These high-speed winds can have significant impacts on weather patterns and the aviation industry.
- In the core of a jet stream (called a Jet Streak), wind speeds are often higher than 100 knots and are occasionally higher than 200 knots.

Polar and Subtropical Jet streams

- The polar jet can sometimes merge with the subtropical jet if it sweeps southward enough, and it occasionally splits into two jet streams.





Thank You

Pillow lava at CIR observed during manned submersible dive